



Figure 1: OpenNebula deployment model

other machines in the cluster, so a public key is used to identify the front-end. All worker nodes need to authorise the front-end SSH key.

After the *opennebula* package has been installed, the next step is to add the worker nodes, but before doing that, you need to install the worker node OpenNebula package on the worker node machines. At both worker nodes, you need to install the *opennebula-node* package by typing the following command:

```
sudo apt-get install opennebula-node
```

This package installs the necessary worker node packages and creates the *oneadmin* user. The following command is used to perform any operation related to the hosts:

```
onehost [options] command [parameters]
```

The command *argument* specifies the operation you want to carry out. You can add new machines to your cluster, retrieve monitoring information, enable or disable scheduling of virtual machines to the host, and list the currently active machines.

The *onehost create* command assists with adding new worker nodes to a cluster, as shown below:

```
onehost create hostname im_mad vmm_mad tm_mad
```

The *hostname* argument is the *hostname* or the IP of the machine; the next three parameters relate to the hypervisor and the transfer manager service. OpenNebula uses different drivers to access different hypervisors. These three parameters essentially tell the front-end that the machine you are adding deploys this specific hypervisor, and this mechanism should be used to transfer data to this machine. In this example, I add a host deploying KVM:

```
onehost add open-nebula-wn im_kvm vmm_kvm tm_ssh
```

The final arguments relate to the way virtual machines will be transferred to the node. In this example, SSH is used to transfer virtual machines. Note that the *oneadmin* user on this host needs to be able to SSH to *oneadmin@open-nebula-wn*; so, on this host, run the following command:

```
sudo -u oneadmin ssh open-nebula-wn
```

Then, verify the host's authenticity. On *open-nebula-wn*, run the following code:

```
sudo apt-get install opennebula-node
sudo tee /var/lib/one/.ssh/authorized_keys << EOT
onehost add open-nebula-wm2 im_kvm vmm_kvm tm_ssh
```

On *open-nebula-wm2*, run:

```
sudo apt-get install opennebula-node
sudo tee /var/lib/one/.ssh/authorized_keys << EOT
```

The *onehost create* command will prompt for the front-end's key to the authorised key file, which OpenNebula uses internally. So, you have to go back to the worker nodes and add the SSH key to these nodes.

If everything goes well, you can check the status of the nodes by typing the *onehost list* command, as follows:

```
01 nebula-user@nebula-cloud-server:~$ onehost list
02 HID NAME      RVM  TCPU  FCPU  ACPU  TMEM  FMEM  STAT
03 0 open-nebula-wm 0 100  99  100 1068948 921356 on
04 1 open-nebula-wm2 0 100  12  100 1173072 1027776 on
```

OpenNebula configuration file

You can specify the configuration for your OpenNebula set-up in the *oned.conf* file. This file is critical to your OpenNebula deployment. The folks at OpenNebula provide all the possible configurations for existing drivers that ship with the standard OpenNebula distribution. You can comment and uncomment as required for your environment. If, for example, you are interested in using Xen, just uncomment the relevant section in the *oned.conf* file as shown below:

```
01 #IM_MAD = [
02 #   name      = "im_xen",
03 #   executable = "one_im_ssh",
04 #   arguments  = "im_xen/im_xen.conf",
05 #   default    = "im_xen/im_xen.conf" ]
06
07 #VM_MAD = [
08 #   name      = "vmm_xen",
09 #   executable = "one_vmm_xen",
10 #   default    = "vmm_xen/vmm_xen.conf",
11 #   type       = "xen" ]
```